

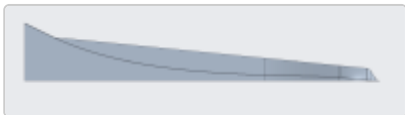
Diffusor

1. KIT23 (concept evaluation)

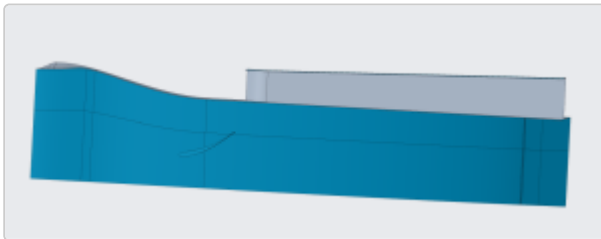
- steady curvature → no „kick“, no curvature change
- curvature is attached to the Heckdeckel and continues after it
- Vortex channel on the sides
- Expansion rearwards parallel to the monocoque in Y-direction

2. Improvements

- strong kick at the front
- curvature change and steady curvature after kick → danger of not enough steepness after the change of curvature



- Vortex generators for vortices in the diffuser →



- Intake for the vortex channel

3. Unsuccessful concepts

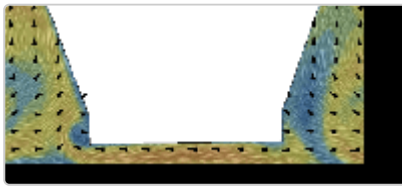
- Diffusor expansion in Y-direction in addition to Z-Expansion → slows down the air too much → air gets sucked upwards and kills the rear wing, does not gain much downforce compared to the baseline
- Rounding on the track side between diffuser and vortex channel → vortex does not release as good on rounding compared to edge
- doubled, smaller VG instead of one big → weaker vortices and more drag

- 3D-curved VG to reduce drag → as expected the Vortex is smaller, but roughly the same flow field. Diffusor slightly gains, RW slightly loses; minimal (0.005) efficiency boost, but 0.1 cl loss → not worth it

4. Conclusion

An improvement to the diffusor helps the whole aerodynamic package in overall downforce production. Therefore, we prioritised updating our diffusor in the development phase.

Because of the low pressure area at the beginning of the diffusor, surrounding air gets sucked into it:



The early kick helps to keep this air within the borders of the diffusor. The air coming from the bottom side of the monocoque, as well as the surrounding air is mostly low energetic in terms of total pressure. In contrast to that, the air just above the ground has a higher total pressure coefficient. To use this air we take the gained space from the kick and place vortex generators after the change of curvature. The created vortices then lead to the attachment of high energetic air on the diffusor surface.

As a result we can produce more downforce and create less counter pressure for other aerodynamic devices.

